AI vs. Human Text Detection

A Data Analyst Deep Dive

Made by **Asma SAIBI**

Kaggle Dataset • 487,235 Essays • Machine Learning Classification

Tools: Python • Pandas • Scikit-learn • XGBoost • Textstat

487,235 Total Essays	**305,797** Human Essays *62.8%*	**181,438** AI Essays *37.2%*	**25+** Features Engineered	**0.992** Best AUC Score

01 | Dataset Overview & Class Distribution

The dataset contains 487,235 student essays sourced from Kaggle, labeled as either human-written or AI-generated. The class distribution shows a moderate imbalance — 62.8% human vs. 37.2% AI — which is relevant for model evaluation strategy.

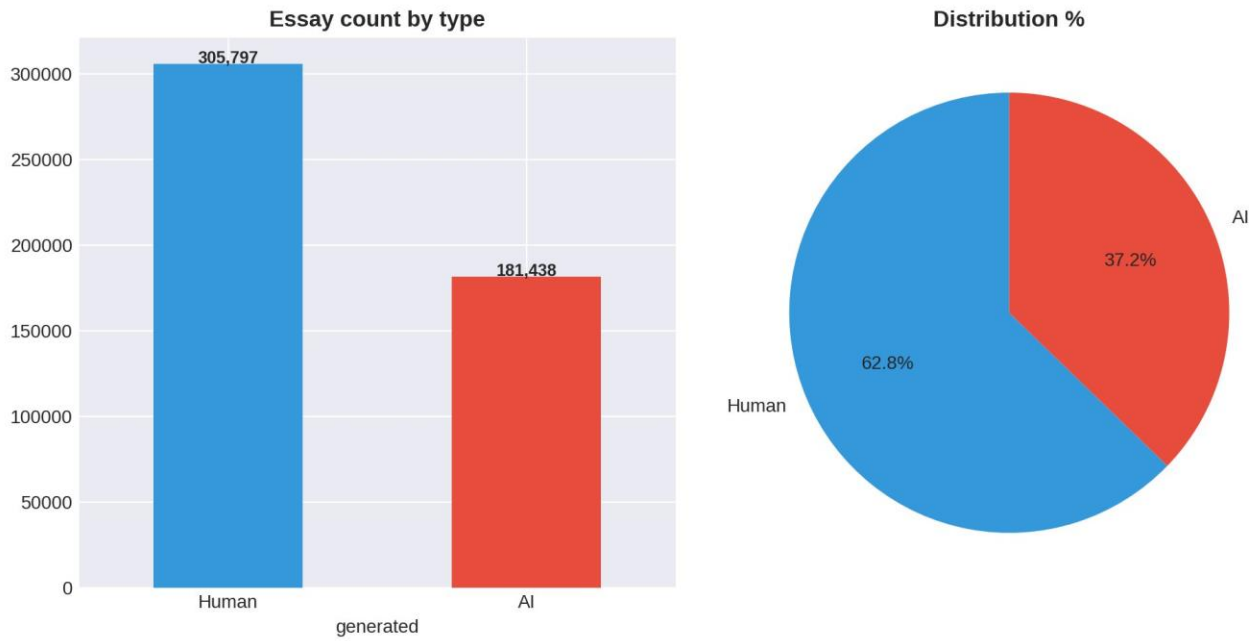

Figure 1 — Essay count by type (bar) and percentage split (pie). Human essays outnumber AI by ~1.69×.

Key Observation

The 37.2% AI share is substantial enough to train robust classifiers. However, class imbalance should be monitored — particularly its effect on recall for the minority (AI) class.

02 | Text Feature Engineering (25+ Features)

Each essay was processed to extract linguistic and structural features across five categories: length/count metrics, vocabulary richness, sentence structure, punctuation patterns, and readability scores.

Key Metrics Comparison

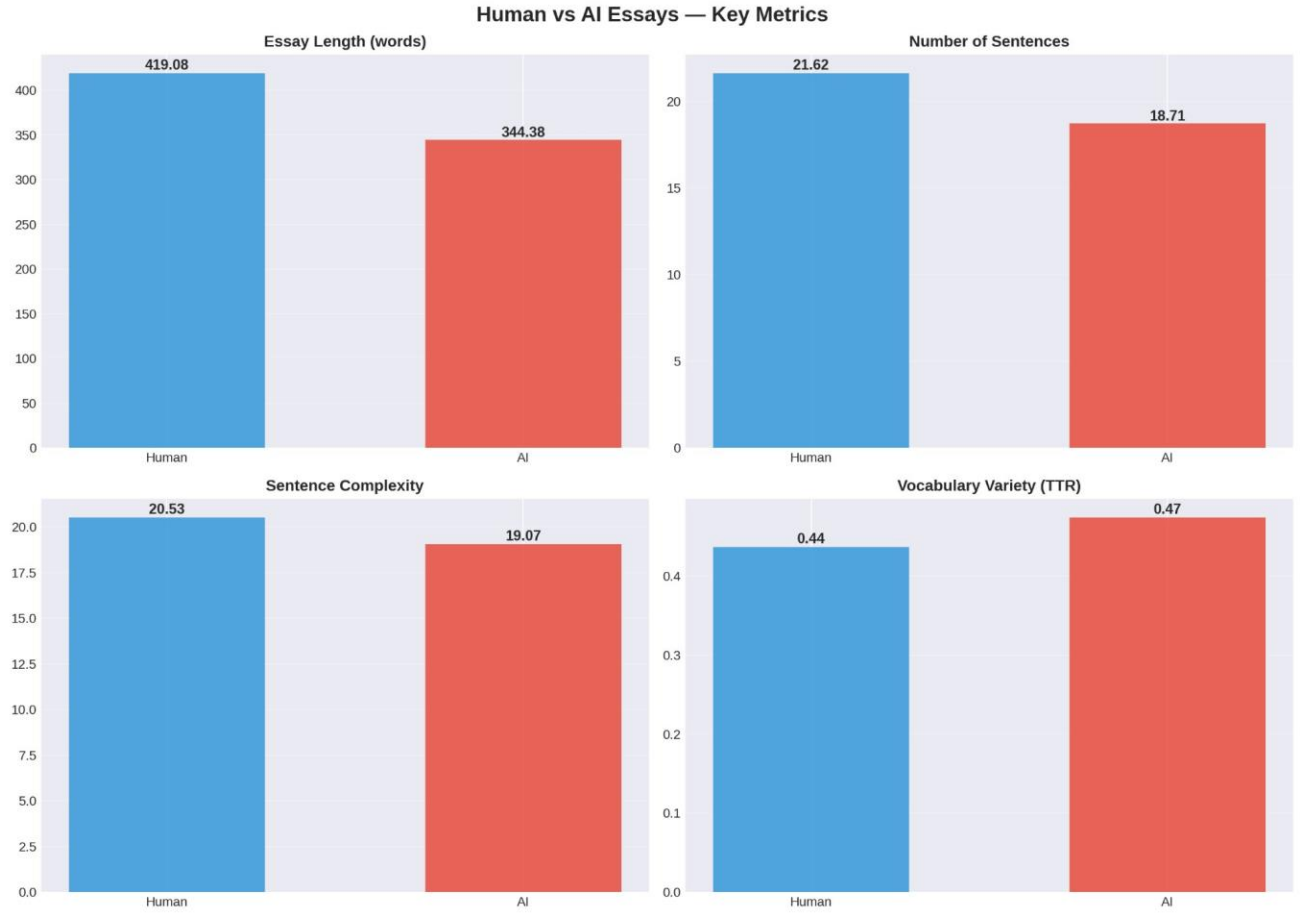

Figure 2 — Human essays average 419 words and 21.6 sentences vs. 344 words and 18.7 sentences for AI. Interestingly, AI scores higher on vocabulary variety (TTR: 0.47 vs 0.44).

Feature Distributions — Box Plots

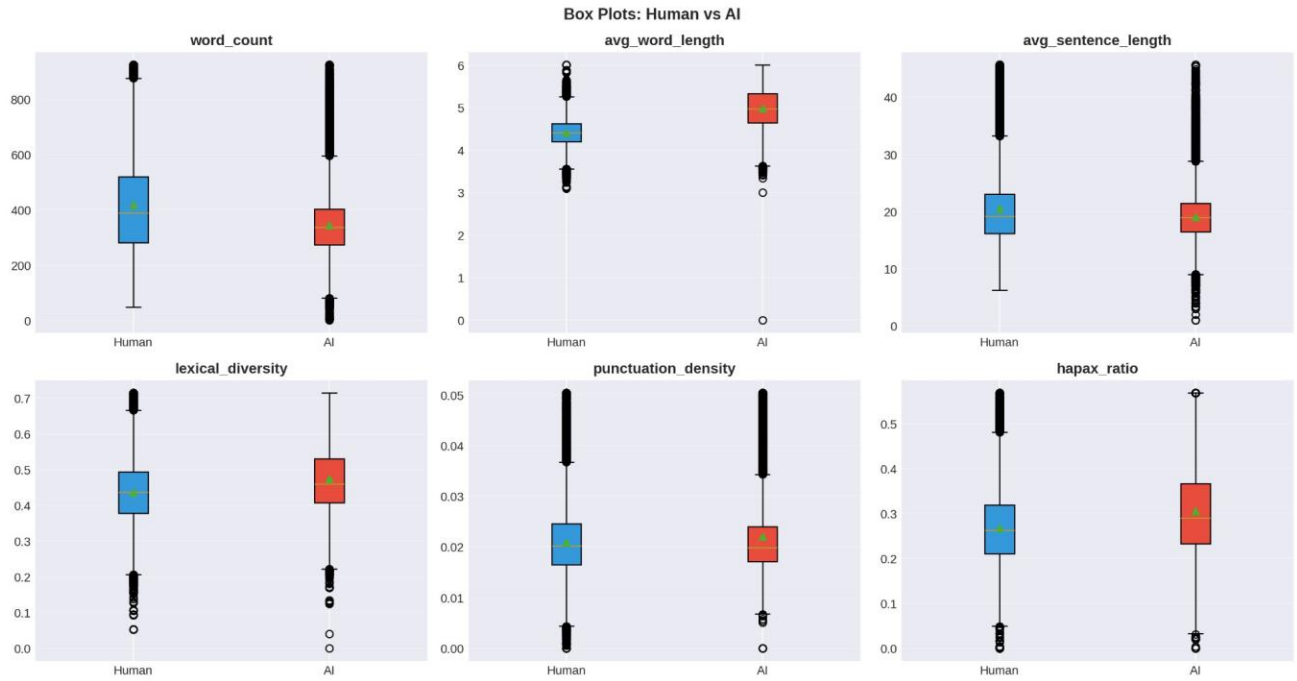

Figure 3 — Box plots for 6 core features. AI text shows higher avg_word_length and lexical_diversity with tighter IQR, indicating more consistent vocabulary use.

Density Distributions

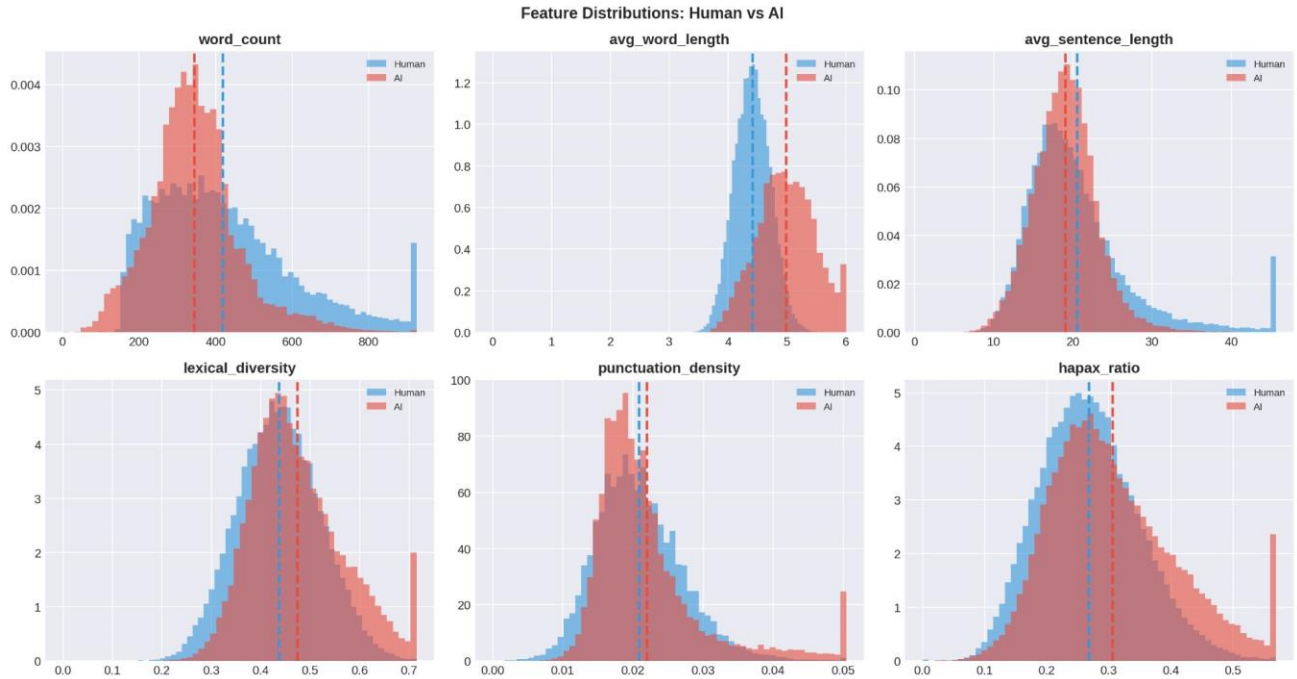

Figure 4 — Density overlays reveal word_count and avg_word_length as the most visually separable features between the two classes.

03 | Readability & Correlation Analysis

Readability Scores: Human vs AI

Figure 5 — AI essays score consistently harder across all readability indices. Flesch Reading Ease: 47 (AI) vs 64 (Human). Flesch-Kincaid Grade: 11.5 (AI) vs 9.5 (Human).

Readability Insight

AI-generated text tends toward more complex vocabulary and longer sentences — resulting in higher grade-level scores and lower ease scores. This is a strong discriminating signal.

Feature Correlation with AI Label

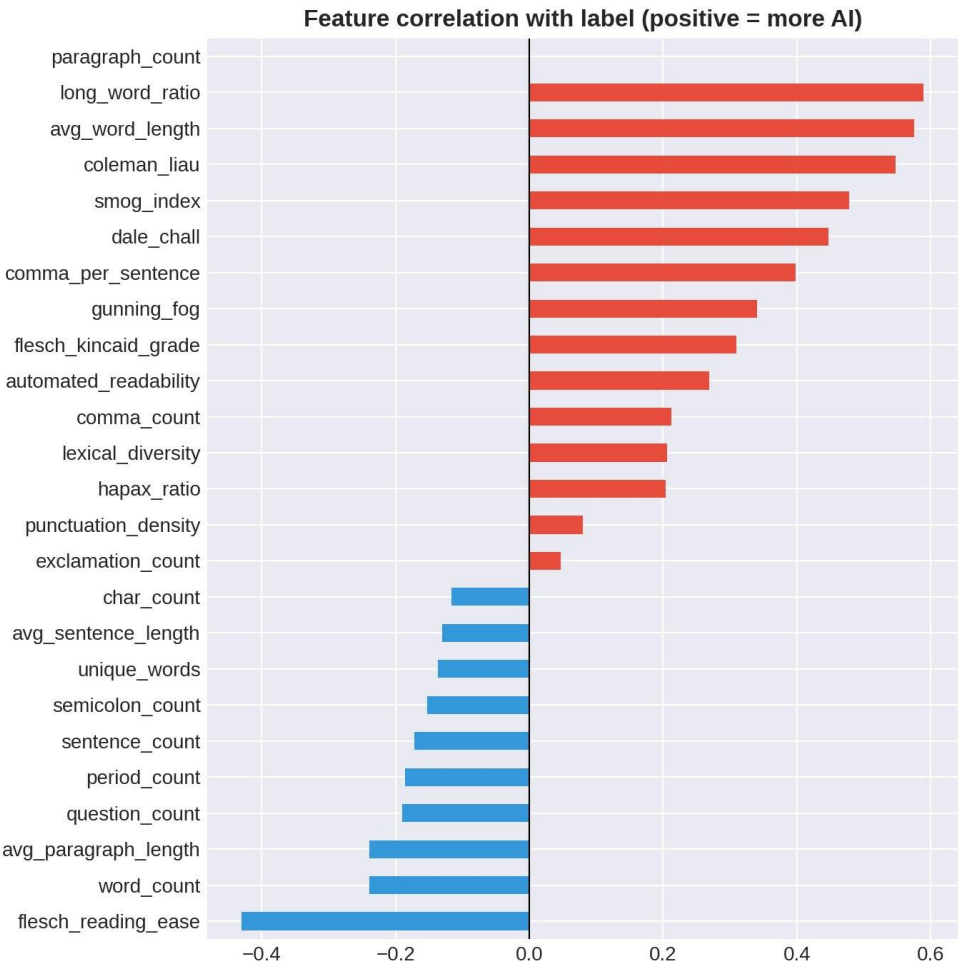

Figure 6 — Feature correlation with label (positive = more AI). long_word_ratio (+0.58), avg_word_length (+0.57), and coleman_liau (+0.54) are the strongest positive predictors.

Features negatively correlated with AI label: flesch_reading_ease (−0.48), word_count (−0.33), avg_paragraph_length (−0.30) — suggesting AI texts are shorter but denser.

04 | Vocabulary Analysis — Top Content Words

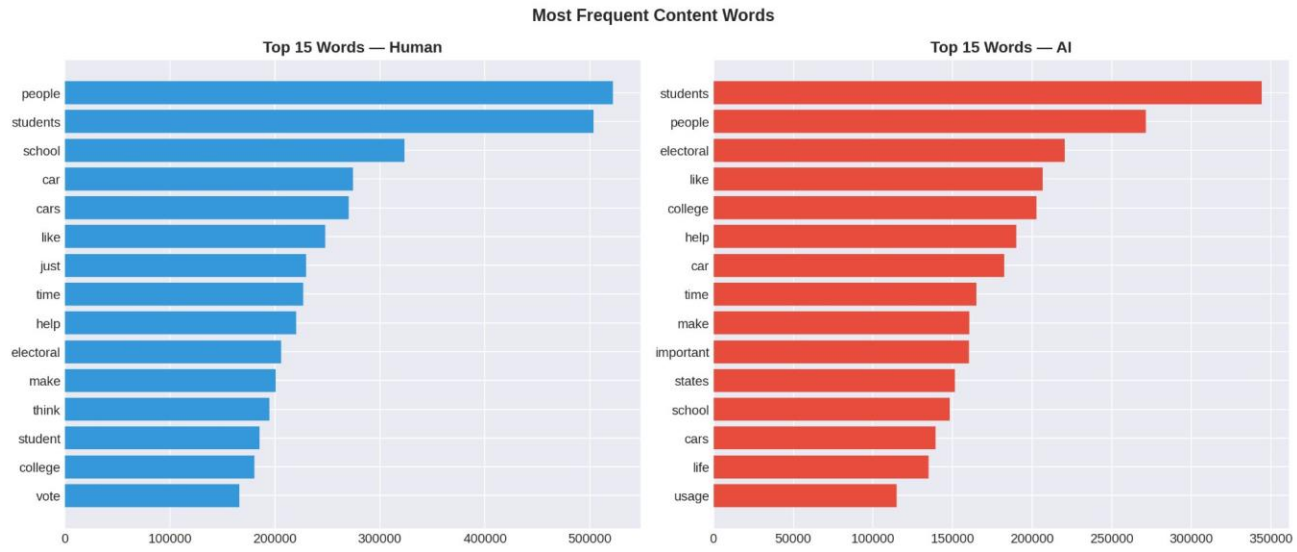

Figure 7 — Top 15 content words for each class. Both share "students", "people", and "school", but AI favors "electoral", "college", "important", and "states" — suggesting more formal, essay-prompt-driven vocabulary.

Vocabulary Pattern

Human essays use casual language ("just", "think", "car/cars") while AI essays lean toward structured argumentative language ("electoral", "important", "states", "usage"). This lexical fingerprint is a meaningful classification signal.

05 | Statistical Significance of Features

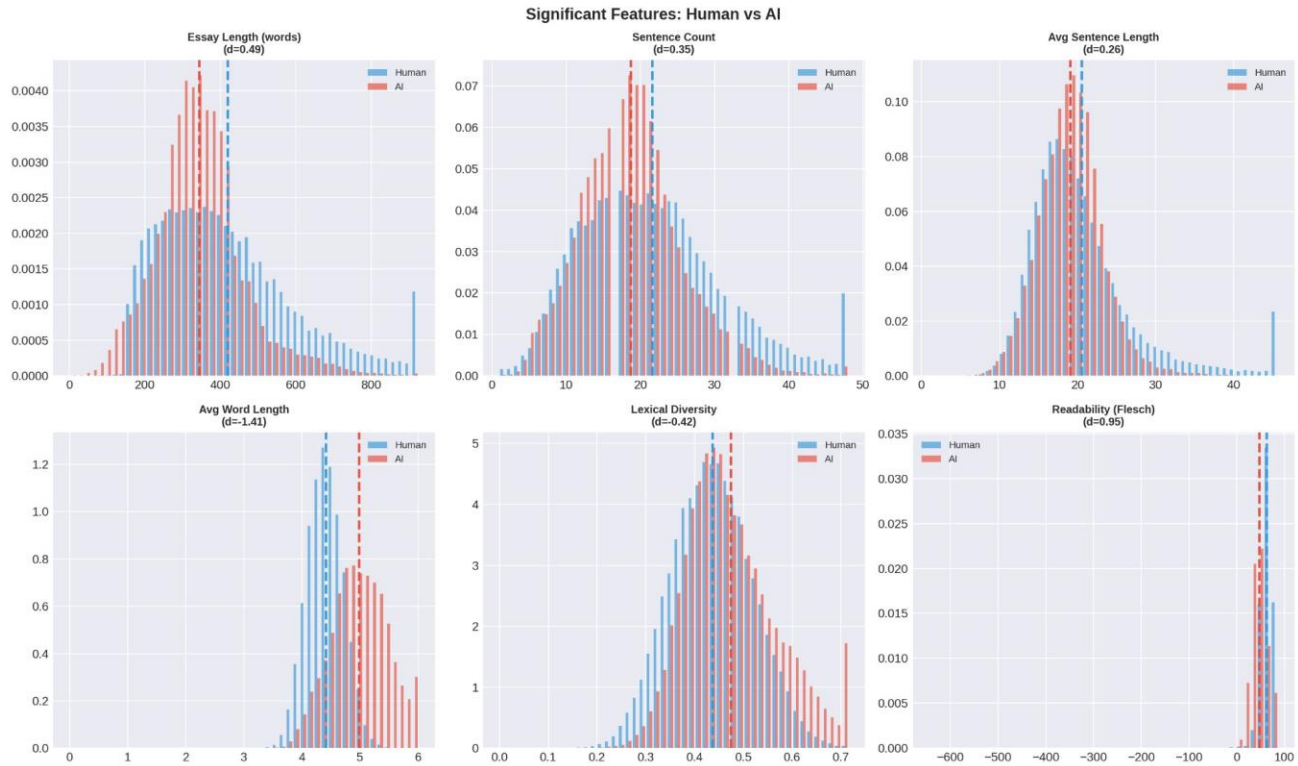

Figure 8 — Cohen's d effect sizes per feature. Readability (Flesch) shows the largest separation (d=−0.95), followed by Avg Word Length (d=−1.41) indicating a large practical difference.

Cohen's d interpretation: |d| < 0.2 = negligible, 0.2–0.5 = small, 0.5–0.8 = medium, > 0.8 = large effect.

Feature	Cohen's d	Effect Size
Avg Word Length	−1.41	Large — AI uses longer words
Readability (Flesch)	−0.95	Large — Human text is easier
Lexical Diversity	−0.42	Small-Medium — AI more diverse
Essay Length (words)	0.49	Small-Medium — Humans write more
Sentence Count	0.35	Small — Humans use more sentences
Avg Sentence Length	0.26	Small — Humans use longer sentences

06 | Machine Learning Classification Results

Three classifiers were trained on the 25 engineered features using a stratified 80/20 train-test split. Results demonstrate that stylometric features alone are highly predictive of authorship.

Logistic Regression	🏆 Random Forest	XGBoost
ACC: 88.0%	**ACC: 95.7%**	**ACC: 94.0%**
AUC: 0.952	AUC: 0.992	AUC: 0.987

ROC Curves

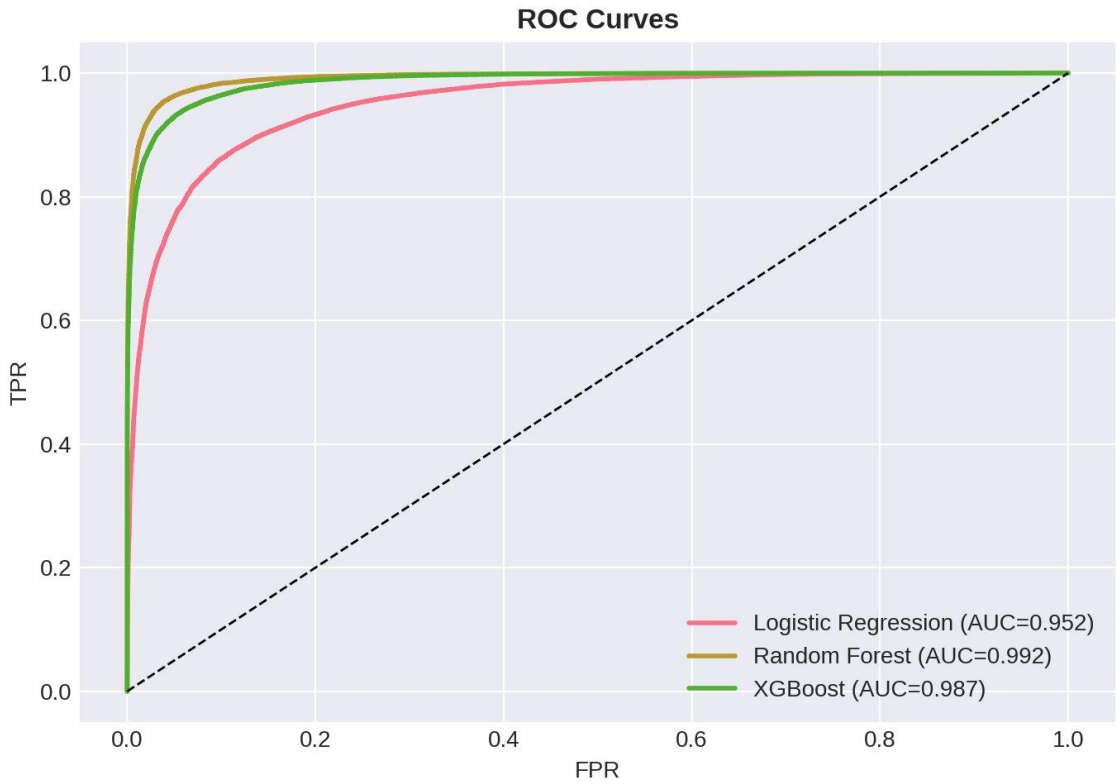

Figure 9 — ROC curves for all three models. Random Forest (AUC=0.992) and XGBoost (AUC=0.987) both achieve near-perfect separation. Logistic Regression (AUC=0.952) is strong but trails behind.

Confusion Matrices

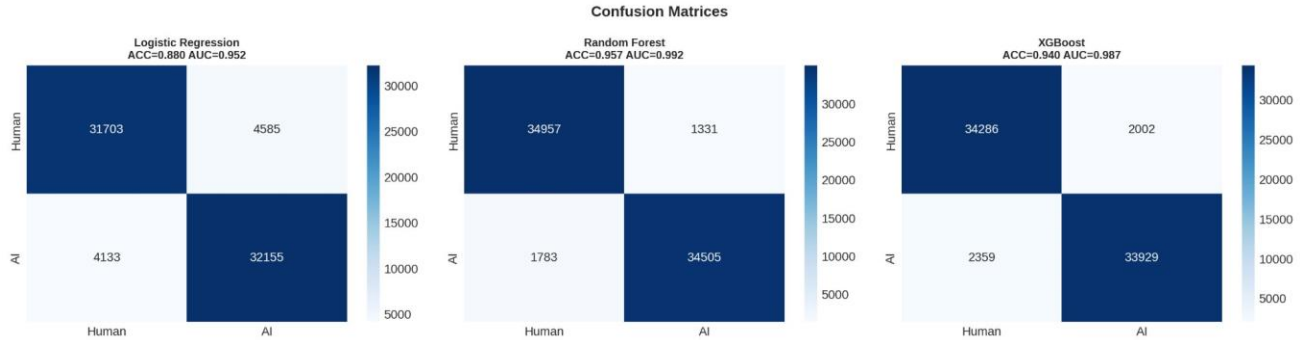

Figure 10 — Confusion matrices. Random Forest misclassifies only 3,114 essays out of 72,576 (4.3% error rate). XGBoost error rate: 6.0%.

Top 20 Feature Importances (Random Forest)

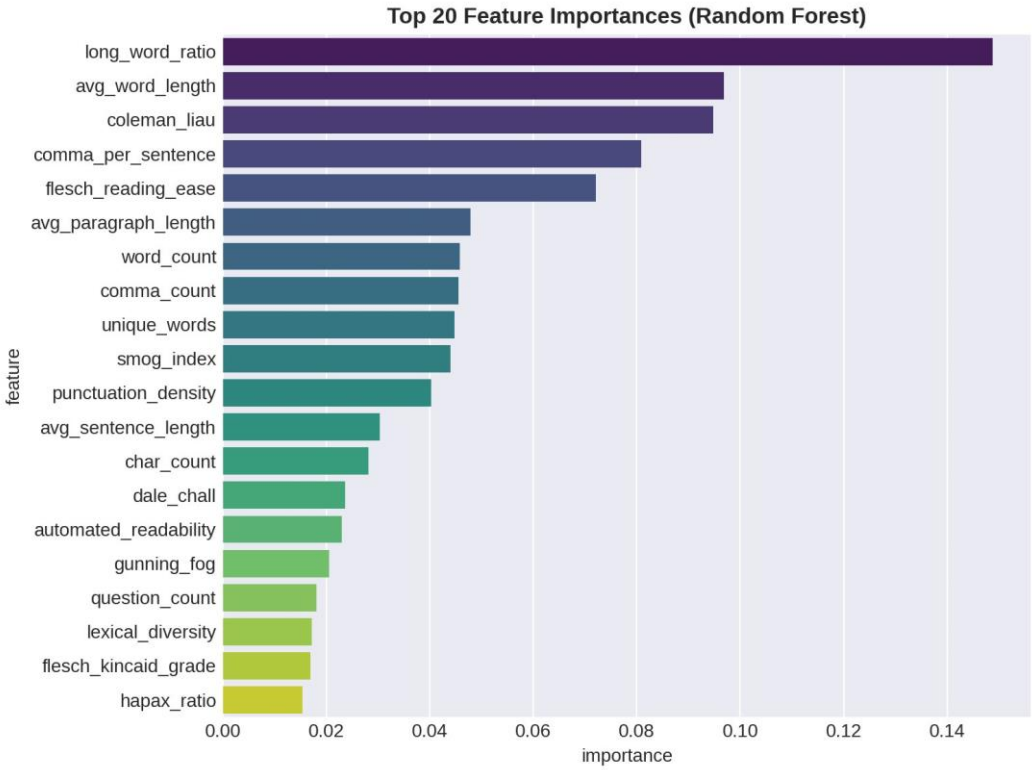

Figure 11 — long_word_ratio, avg_word_length, and coleman_liau dominate feature importance, confirming that vocabulary complexity is the primary classifier signal.

07 | Key Findings & Business Takeaways

What Makes AI Text Detectable?

- AI uses longer, more complex words — long_word_ratio and avg_word_length are the #1 and #2 features
- AI text scores harder on all readability indices (Flesch, Coleman-Liau, SMOG, Gunning Fog)
- Human essays are longer on average (419 vs 344 words) with more sentences
- AI vocabulary is more diverse (higher TTR) but also more formal and prompt-focused
- Punctuation density and comma_per_sentence are moderate but consistent AI signals

Model Performance Summary

- Random Forest achieves 95.7% accuracy and AUC=0.992 — best overall model
- XGBoost achieves 94.0% accuracy and AUC=0.987 — strong alternative with faster inference
- Even Logistic Regression reaches 88% accuracy, confirming features are linearly separable
- All models significantly outperform the 62.8% baseline (majority class classifier)

Strategic Recommendation

Deploy Random Forest as the production classifier for AI text detection. Focus monitoring on long_word_ratio, avg_word_length, and coleman_liau as the top-3 real-time features. Consider periodic retraining as AI writing models evolve and their stylometric fingerprints shift over time.

Analysis performed with Python • Pandas • Scikit-learn • XGBoost • Textstat

Dataset: Kaggle — Distinguishing AI-Generated from Human-Written Text